

CO1-AT1 - short answer Test

Name: Vyshnavi Gk.

Reg. No: 192425069

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Course Name: Computer Networks.

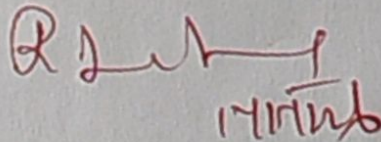
Faculty Name: Dr. R. Senthamil Selvan.

Section, Batch: AIm1 - 2024

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Total Marks: 20 Marks

Obtained marks: 20

Faculty incharge: 
R. Senthamil Selvan

course coordinator:

Hod:

File Transfer (Transport & Data link layer)

Given

$$\text{File size} = 150 \text{ MB} = 150 \times 10^6 = 150,000,000 \text{ bytes}$$

$$\text{Segment size} = 1500 \text{ bytes}$$

$$\text{Header + Trailer} = 40 \text{ bytes/frame}$$

$$\text{Band width} = 50 \text{ mbps}$$

Calculation:

$$\text{No. of segments: } 150,000,000 \div 1500$$

$$= 100,000 \text{ segments}$$

$$\text{No. of frames: } 10,000 \text{ frames}$$

$$\text{Total overhead: } 100,000 \times 40$$

$$= 4,000,000 \text{ bytes (4 MB)}$$

$$\text{Total data transmitted: } 150 \text{ MB} + 4 \text{ MB}$$

$$= 154 \text{ MB}$$

$$\text{Convert to bits: } 154 \times 8 = 1232 \text{ MB}$$

$$\text{Transmission time: } 1232 \div 50$$

$$= 24.64 \text{ sec}$$

Explanation:

1. Transport layer divides data into segments and ensures reliable delivery using acknowledgements and error recovery.
2. Data link layer creates frame, detects errors using CRC.

Sliding window:

Given

Window size = 6

Total frames = 48

Calculation:

$$\text{Windows required: } 48 \div 6 = 8$$

one frame lost in each window

Retransmissions = 8

Flow control prevents sender overflow by matching transmissions speed with receiver

IP Fragmentation:

Packet = 12,000 bytes

MTU = 1500 bytes

IP header = 20 bytes

etc calculation:

payload per fragment: $1500 - 20$

= 1480 bytes.

Fragments required = ceiling $(12000 \div 1480)$

= 9 fragments.

Header overhead:

$$9 \times 20$$

= 180 bytes.

sliding window.

Given:

Window size = 6

Total frames = 42

$$\text{windows required} = 42 \div 6 \\ = 7$$

one frame lost in each window.

Retransmissions: 2

Explanation:

Flow control prevents sender overflow by matching transmission speed with receiver capacity.

5. session layer

Given:

File = 600mb

check point every 60mb

Failure after 390mb.

Calculation:

$$\text{check points before failure} = 390 \div 60 \\ = 6$$

$$\text{last check point} = 360 \text{ mb}$$

$$\text{Data to retransmit: } 390 - 360 \\ = 30 \text{ mb}$$

Explanation:

Session layer synchronization check points allow transmission to resume from the last check point.

6. presentation layer.

Given:

original file = 900mb

Compression = 40%.

Encryption overhead = 5%.

Calculation:

Compressed size = $900 \times 60\% = 540\text{mb}$

Encrypted size = 567mb

Reduction = $900 - 567$

= 333mb

Percentage reduction: $(333 \div 900) \times 100$

= 37%.

Explanation:

Present layer compresses data to reduce transmission time and encrypts.

7. FTP Application:

Given:

Data = 3 GB = 3000m

Request size = 3mb

Throughput = 120mbps.

Calculation:

Requests = $3000 \div 3$

= 1000

Data in bits: 3GB = 24,000mb

Transmission line: $24000 \div 120$

= 200s.

End to End delay

Given

Processing delay

Application = 4ms

Transport = 3ms

Network = 5ms

Data link = 2ms

Total processing: 15ms

Total delay: $15 + 18 + 12$

$= 45\text{ms}$

Explanation: Each OSI layer performs specific processing before transmission. Propagation delay depends on distance.

9. protocol overhead:

useful data = 80mb

Frame = 1600 bytes

overhead = 80 bytes

Calculation:

useful payload = $1600 - 80 = 1520$ bytes

Frames required: $80,000,000 \div 1520$
 $= 52632$ frames.

Total overhead: 4,210,560 bytes

$= 4.21\text{mb}$

Efficiency: $(1520 \div 1600) \times 100$

$= 95\%$

Explanation:

Protocol overhead increases transmitted data and slightly reduces network efficiently.

10. Healthcare network.

Given:

original data = 240mb = 240,000,000 bytes

compression = 30%.

Segment = 1400 bytes

Calculation:

compressed size = 240×70

= 168mb

Segments = $168,000,000 \div 1400 = 120,000$ segments

protocol overhead = $120,000 \times 60 = 7.2$ mb

Final transmitted data: $168 + 7.2 = 175.2$ mb

Explanation:

presentation layer compresses and encrypts data

Datalink layer add frame headers/trailers and perform error detection.

understanding Concepts	explanation	Application	Organization	Accuracy	Clarity
25%.	25%.	20%.	15%.	10%.	5%.
5	5	4	3	2	1